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ORIGINAL ARTICLE

Comparison of four machine learning algorithms for a pre-impact fall detection system

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Abstract
In recent years, real-time health monitoring using wearable sensors has been an active area of research. This paper presents an efficient and low-cost fall detection system based on a pair of shoes equipped with inertial sensors and plantar pressure sensors. In addition, four machine learning algorithms (KNN, SVM, RF, and BP neural network) are compared in terms of their detection performance and suitability for pre-impact fall detection. The results show that KNN and BP neural network outperformed the other two algorithms, where KNN had 98.8% sensitivity, 99.8% specificity, and 99.7% accuracy, and BP neural network had 100% sensitivity, 99.8% specificity, and 99.9% accuracy. KNN outperformed BP neural network in terms of fitting ability, and their lead times were both 460.95 ms. The system can provide sufficient intervention time for the wearer in the pre-impact phase and together with the touchdown fall protection device can effectively prevent fall injuries.

Keywords Fall detection · The elderly · Wearable · Multisensor · Pre-impact

1 Introduction

At present, population aging is an important feature of global demographic trends, and it is becoming increasingly serious [1]. According to the World Health Organization (WHO) [2], the frequency of falls increases with age and frailty. More than one-third of people over the age of 65 fall more than once a year, and more than two-thirds of people over the age of 85 fall twice or more per year [3]. Falls can cause serious health problems in the elderly, such as fractures, joint sprains [4, 5], and even death [6]. Even if a fall does not result in injury, it can create a fear of falling in older people that threatens their ability to live independently. Reducing the impact of falls on the elderly is therefore the key to achieving healthy aging. It is necessary to detect falls as early as possible and allow more time for fall intervention to reduce the risk of injury in the elderly [7].

Existing fall detection is divided into post-impact detection [8] and pre-impact detection [9]. Although post-impact detection systems achieve high detection accuracy and will provide timely medical assistance to fall victims, thus avoiding unnecessary losses due to "long-lie" [10], this approach does not reduce the harm caused by impacts to older people. In contrast to post-impact fall detection, pre-impact fall detection aims at predicting fall events prior to impact [11]. If a fall is predicted at the earliest stage and there is sufficient time (advance response time) to implement protective interventions, such as the use of airbag systems [12] or inflatable hip protectors [13], the damage will be greatly reduced. One of the most important parameters in terms of pre-impact fall detection is considered to be the pre-impact lead time, which is defined by the time interval between the detection of a fall and the fall impact. And it takes into account the time to activate protective measures to protect the fall victim from the fall impact [14]. Therefore, the longer the lead time, the better the fall detection performance. Shi et al. [15] proposed an inflatable hip proctor to prevent fall impact. The inflation time of an airbag is reported to be approximately 130 ms. Therefore, for our experimental design to effectively prevent fall impacts, the lead time should be longer than 130 ms.

Previous research on pre-impact fall detection can generally be divided into three categories: vision-based, environmental perception-based, and wearable device-based [16].

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The vision-based technology [17–21] establishes a motion capture system by the cameras installed in fixed positions, which can accurately determine the fall detection indicators and has a high recognition accuracy. There are also some studies that combine HAR technology to detect falls and trigger corresponding alarms by recognizing human movements [22]. However, the implementation of a motion capture system is limited by the application space, because the amount of capture is always limited, and it is likely to expose the user's privacy.

The fall detection system based on environmental perception [23–25] deploys one or more ambient sensors (such as microphones and infrared and pressure sensors) in the human activity areas to collect information on human movement and posture to determine whether a fall has occurred. It does not need to be worn and has little impact on the user's activities. However, it is more expensive than the wearable systems and can only be used indoors with a relatively high false alarm rate.

Most existing pre-impact fall detection systems are based on wearable sensors (e.g., accelerometers, gyroscopes, and pressure sensors). They can collect measurements and classify user actions into falls and activities of daily living (ADL). Small size, light weight, and low cost allow wearable sensors to unobtrusively capture body movements for longer periods of time, as well as monitor kinematic measurements over time. This makes them suitable for fall detection before impact. Compared to vision-based technology and environmental perception systems, wearable sensors have the advantage of being compatible with various physical environments, are not limited by space, and can easily communicate with smartphones [26] or other Internet-enabled devices [27]. In addition, wearable devices do not invade the user's privacy during the normal fall detection process, and the cost is lower than the other two detection methods.

Over the last decade, many wearable-based pre-impact detection studies have been conducted. A wearable pre-impact fall detector using feature selection and support vector machines has been proposed in the literature [28]. They used a wearable three-axis Micro Electro-Mechanical Systems (MEMS) accelerometer to collect data from human motion information, and the system they implemented was able to detect falls on average 203 ms before impact. Lee et al. [29] proposed a vertical velocity-based pre-impact fall detection method using wearable inertial sensors. Liang et al. [30] proposed an HMM-based SVM classifier to determine falls and activities of daily living (ADLs) in a wearable sensor detection system. Some studies [13, 31, 32] have used inertial measurement units (IMUs) consisting of triaxial accelerometers and triaxial gyroscopes to design pre-impact fall detection systems. For example, Soonjae [33] and Zhong [34] have proposed the use of waist-worn IMUs for fall detection based on the threshold method. However, the limited data information from individual sensors often results in a high false alarm rate, and in the study by [14], the specificity of the algorithm was only 78.3%–83.9%, despite achieving 100% sensitivity in detecting falls.

To overcome these limitations, several researchers have attempted to use multiple sensors to detect falls [35]. Sabatini et al. [36] combined inertial sensors with a barometric altimeter (BIMU) to determine the change in vertical speed and height during a fall. Their results showed a specificity of 100% and sensitivity of 90%, with a mean time before impact of 157 ms. In the literature [32], a model for pre-impact fall detection based on discriminant analysis has been presented using two inertial sensors mounted at the waist and thigh. Reference [37] proposed a threshold-based fall detection method based on the use of a pair of instrumented insoles as a wearable device. Three force-sensing resistors (FSRs) on each insole and a three-axis accelerometer provide the raw data to run a threshold-based algorithm to detect falls with acceptable accuracy and reduced computational complexity. Results show that multisensor fusion can improve recognition accuracy and fault tolerance [35]. The location of the wearable sensors used is an important factor to consider [38]. The results of Martelli et al. [36], which identified the best subset of body parts for fall detection, showed that only the kinematics of the distal upper and lower extremities need to be observed to achieve reasonable performance for effective fall detection.

Since foot motion is critical in falls [39], we develop a multisensor-based wearable fall detection device with a sensor system consisting of a pair of instrumented insoles and an IMU, which is capable of accurately detecting different types of falls and ADLs. A thorough comparison of the performance of four popular machine learning classifiers (k-nearest neighbor (KNN), random forest (RF), support vector machine (SVM), and back propagation (BP) neural network) in detecting falls and identifying different fall activities was conducted. The main contributions of this research are as follows:

- A multisensor-based pre-impact detection system that can be embedded in shoes was developed to achieve an ultra-long lead time of 460.95 ms, which provides new opportunities to reduce the impact risk caused by falls in practical applications.
- The use of a six-degree-of-freedom perturbation platform to simulate the real fall of subjects provides a new experimental method to fall risk study.
- The performance of four common machine learning algorithms (KNN, RF, SVM, and BP) in predicting falls was compared to determine which algorithms are more applicable for fall detection.

The paper is organized as follows. Section 2 describes the wearable pre-impact fall detection device and the pre-processed data. Then, in Sect. 3, a preliminary evaluation test is presented and discussed. Finally, Sect. 4 concludes this article.

2 Materials and methods

2.1 Data acquisition

The data acquisition in this paper is based on a self-designed system. The hardware of the system is mounted on the foot, in a left–right symmetric configuration. It includes two MPU-6050 (InvenSense, San Diego, USA) inertial sensors with an integrated 3-axis MEMS accelerometer and a MEMS gyroscope, which have a measuring range of ± 8 g and ± 2000 dps, respectively. The coordinate system of the inertial sensor is shown in Fig. 1a for the left foot. The smart insoles are each equipped with four FRS402 membrane pressure sensors (Interlink Electronics, Nevada, USA) with an outer diameter of 18 mm and a measuring range of 20 g to 6 kg. The optimal position of the sensor on the insole is crucial as it affects the clear and reliable detection of the gait cycle phase. A preliminary survey suggests that the thumb, first metatarsal, fourth metatarsal, and heel are the most representative positions during human movement [40], as shown in Fig. 1b. The communication module uses the ESP8266 chip (Espressif, Shanghai, China) to implement data transmission, and the baud rate is set to 115,200. The two microcontrollers (MCUs) use STM32F103C8T6 (ST, Switzerland), and the two power modules use 7.4 V lithium batteries to power the entire system. The circuit board integrated with the WIFI module, inertial measurement units (IMUs), and controller is placed inside the 3D printed shell shown in Fig. 1c, which can be fixed to the outside of the shoe. The overall hardware representation of the system is shown in Fig. 1.

The software system of the system is implemented on the basis of PyQt5 graphic editing tool and developed using the Python language. The designed host computer system includes control interface and data display interface. The raw sensor data is transmitted to the host computer via WIFI module and then displayed in real time and stored on the computer.

2.2 Feature extraction

Human behavioral activity is complex and diverse, to achieve the purpose of distinguishing falls from various ADLs, and to improve the accuracy of pre-impact recognition, feature selection of sensing data samples is required to enable correct labeling of falls and ADLs. The raw sensor data collected by the system for a single foot has ten as shown in Eq. (1), where $[A_{ccx}, A_{ccy}, A_{ccz}]$ indicates the 3-axis acceleration values, $[G_{rvx}, G_{rvy,} G_{rvz}]$ denotes the 3-axis angular velocity values, and $[F_1, F_2, F_3, F_4]$ represents the pressure sensor value of the first toe, the first metatarsal, the fourth metatarsal, and the heel, respectively. Since there are two sensing systems installed on the left and right feet, the sensor data is collected for 20 values simultaneously.

$$D = [A_{ccx}, A_{ccy}, A_{ccz}, G_{rvx}, G_{rvy,} G_{rvz}, F_1, F_2, F_3, F_4] \quad (1)$$

Feature selection is performed on data samples to achieve correct labeling of falls and ADLs. First, the zero point of a fall is determined, and then, the first five points including this point are selected to be considered as a fall. Figure 2 shows the feature data extraction for the four types of falls. The threshold points for these four types of falls can be determined by looking at the change in plantar pressure at specific locations. The points included in the entity matrix in the figure are all considered to be fall data points, where the critical point is the point on the far right of the matrix, and the critical position of each fall is determined as follows:

(a) Forward fall: the highest point of left thumb pressure F1_l and right thumb pressure F1_r
(b) Backward fall: the highest point of left heel pressure F4_l and right heel pressure F4_r
(c) Leftward fall: the highest point of the fourth metatarsal pressure F3_l on the left foot and the lowest point of the second metatarsal pressure F2_r on the right foot
(d) Rightward fall: the highest point of the fourth metatarsal pressure F3_r on the right foot and the lowest point of the second metatarsal pressure F2_l on the left foot

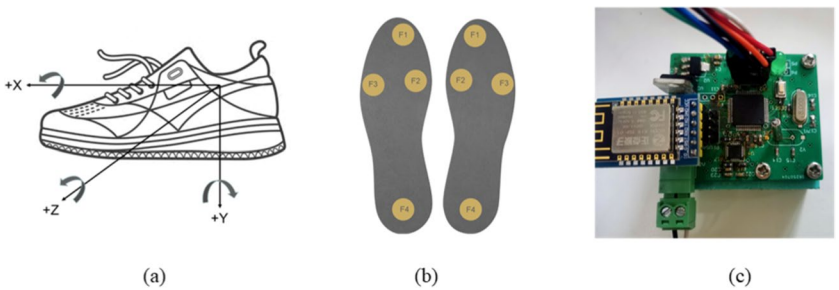


Fig. 1 Hardware part of the system, (**a**) shows the inertial sensor coordinate system of the left foot, (**b**) shows the position distribution of the pressure sensor, and (**c**) shows the physical map of the hardware system

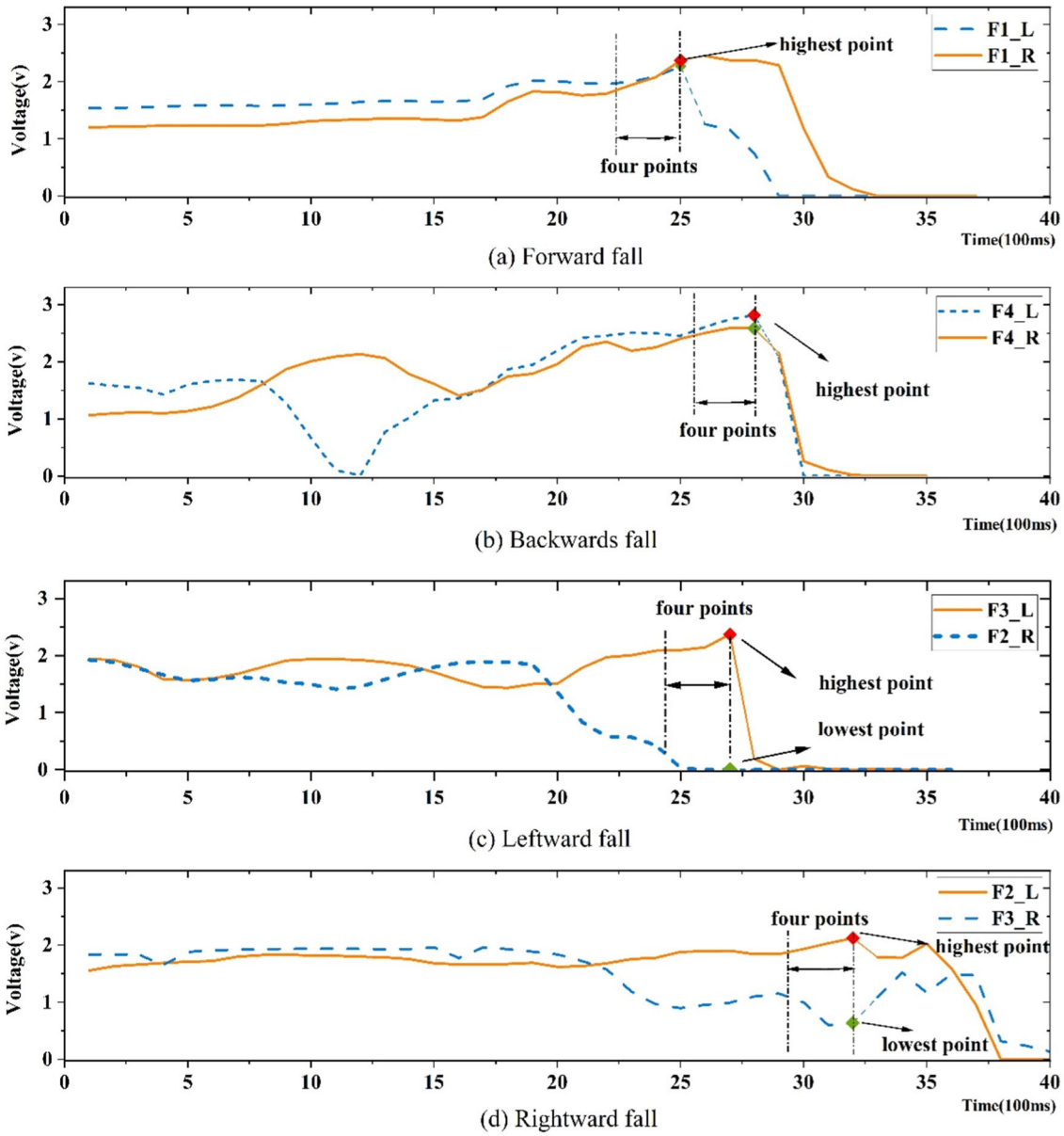


Fig. 2 Feature data extraction for four types of falls

2.3 Data processing

Falling is a dynamic process, and it is difficult to fully reflect this process using data from only one fixed point in time. To solve this problem, a sliding window algorithm is used to further process the data. The sliding window is shown in Fig. 3, where the rightmost part of the rectangular window represents the sensor value at the current point in time. The sliding window length of this system is set to 4, and the average value of the sensor data within the sliding window is used as the training data for the algorithm, as shown in Eq. (2).

$$\overline{X} = \frac{1}{4}\sum_{i=1}^{4} X_i \tag{2}$$

The sampling frequency of this system is 10 Hz, which means that every 100 ms the sliding window advances one frame and the average of the sensor data over 400 ms is calculated and fed into the algorithm to determine the current human posture. As shown in Fig. 3, the sliding window moves to the right along the arrows and only the pressure data is displayed for viewing purposes.

Each sensor data has different range units, so as to eliminate the influence between the dimensions and also

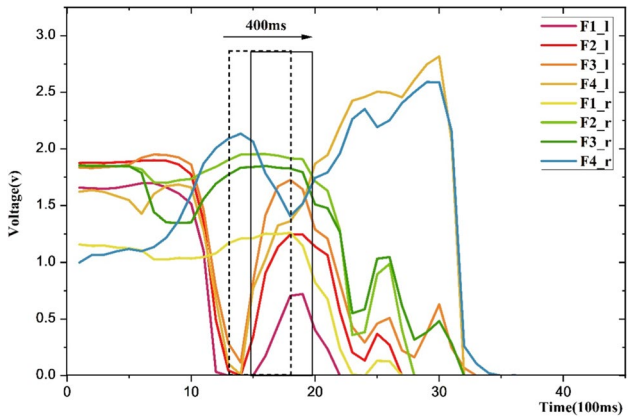


Fig. 3 Sliding window of the acquisition system

to avoid oversized data to affect the convergence of the algorithm model, it is also essential to normalize the data. The system uses the sensor range normalization, each sensor data is divided by its range so that the original data is linearly transformed to map the result between [0, 1]. The transformation formula is shown in Eq. (3), where M is the range of the sensor.

$$X^* = \frac{X}{M} \tag{3}$$

In addition, to ensure the accuracy of the model and to prevent the effect of uneven data distribution on the training results of the algorithm, the collected data must be oversampled. The SMOTE [41, 42] algorithm is used here, and its core is to synthesize new minority class samples as follows. (1) For each sample x in the minority class, calculate the distance from it to all samples in the minority class sample set based on the Euclidean distance, to get its k nearest neighbors. (2) Set the oversampling rate N according to the sample imbalance ratio, and randomly select N samples among its k nearest neighbor samples, which are respectively recorded as $\times 1, \times 2, \ldots, xn$; 3). Then, in the minority class samples, perform random linear interpolation between x and xi ($i = 1, 2, \ldots, N$). Calculate and generate a new synthetic minority class sample $x_{\text{new_}i}$ by Eq. (4):

$$x_{new_1} = \mathrm{x} + \mathrm{rand}(0, 1) * (\mathrm{xi} - \mathrm{x}), \mathrm{i} = 1, 2, \ldots, \mathrm{N} \tag{4}$$

where rand (0, 1) represents any decimal between 0 and 1, and the newly constructed sample is any point between the selected xi and the original sample x. After oversampling in this paper, the ratio of the number of fall data to ADL data is 1:1. In summary, the overall flowchart of the pre-impact fall detection system proposed in this paper is shown in Fig. 4.

3 Experiment and results

3.1 Experimental protocol

To test the performance of the proposed system, an experimental system was set up. As shown in Fig. 5a, the experimental site is located in a laboratory. The experimental equipment includes the pre-impact detection device as shown in Fig. 5b and the six-degree-of-freedom perturbation platform (China, Shanghai, Shanghai Yinghao Electromechanical Equipment Co., Ltd.) to create a falling environment. The six-degree-of-freedom Stewart platform [43, 44] used in this paper mainly includes an upper moving platform, six outriggers, and a lower fixed platform. The outriggers can independently perform telescopic movements and are connected to the upper moving platform and the lower fixed platform by Hooke hinges (or spherical hinges). The upper moving platform has 3 degrees of freedom in translation and 3 degrees of freedom in rotation and can perform 3 translational movements (lateral, longitudinal, and vertical) and 3 rotational movements (pitch, roll, and yaw). The assembly field (for example, aircraft assembly) has high requirements for the positioning accuracy of the Stewart platform, but in this paper, the platform positioning accuracy ≤ 0.2 mm in the balance disturbance experiment can meet the requirements. Under the guidance of professionals, the subjects were trained to simulate an elderly person's fall, and pre-experiments were conducted to determine the offset parameters for setting up the six-degree-of-freedom perturbation platform at 10 mm displacement and 1000 mm/s.

For the reason of safety and efficiency, five healthy subjects with a mean age of 23.4 ± 1.3 years (one female and four males) were recruited for this study. Prior to participation in the experiment, all subjects gave written consent to the experimental protocol, which was approved by the Ethical Review Board of Huashan Hospital affiliated to Fudan University (HIRB No. 2019–059). For the fall category, to ensure that this experimental procedure is carried out as close as possible to actual falls in the elderly [45], five different kinds of ADLs and four directional falls were assessed including (a) walking: as steady a speed as possible, (b) standing, (c) walking upstairs: in the real staircase, (d) walking downstairs, (e) standing up and sitting down, (f) forward fall, (g) backward fall, (h) leftward fall, and (l) rightward fall. All activities were timed to be completed within 10 s, and stair climbing and walking were kept at a constant speed as much as possible. Each activity was performed 4 times by 5 subjects.

The system works with a sampling frequency set at 10 Hz. After excluding some invalid data due to device communication and the feature extraction and data processing mentioned in Chapter 2, the final 84 fall data

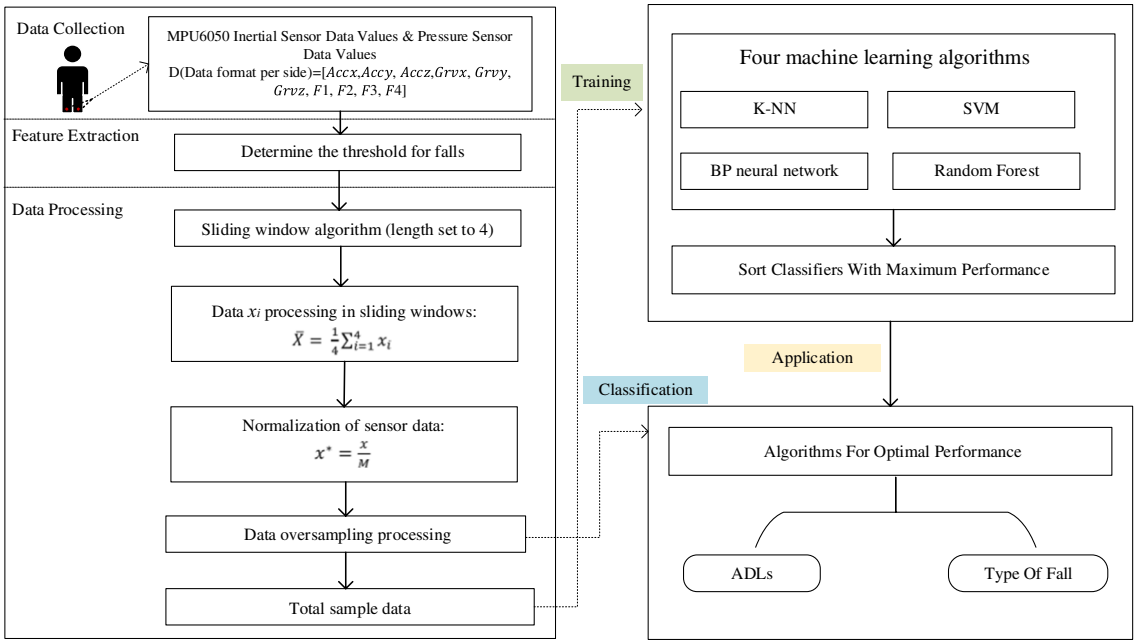


Fig. 4 The proposed falling detection system

points and 1940 ADL data points were used as training samples to train the four algorithms. The distribution of numbers by activity is shown in Table 1.

3.2 Algorithm performance assessment

In order to realize an efficient fall detection system, this paper presents a comparative analysis of four commonly used machine learning algorithms, including BP neural network, KNN, SVM, and RF. The four algorithms are implemented using the Scikit-learn library in Python and are set up as shown below.

(1) BP neural network: in this paper, a three-layer BP neural network is designed. To choose MLPClassifier as the classification model, the hyperparameters that need to be set include hidden_layer_sizes, max_iter, and solver. Among them, the grid search algorithm [46] in the Scikit-learn library is used for the number of hidden layer neurons hidden_layer_sizes and the maximum number of iterations max_iter, combined with cross-validation to obtain the optimal hyperparameter values of 10 and 1000, respectively. Random state is set to 1 by default. The solver has three built-in parameters: lbfgs, sgd, and adam. Among them, the random stochastic gradient optimizer adam performs well on relatively

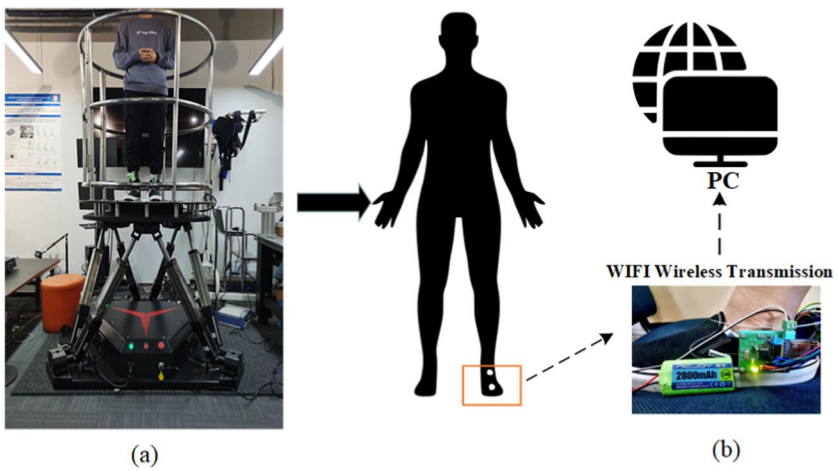


Fig. 5 The process of the fall test, (**a**) shows the physical picture of the test system used in the test, and (**b**) shows the schematic diagram of wearing the pre-collision detection device

Table 1 The amount of data collected by each activity

Experimental type	Experimental trials	Collection quantity	Total amount
Fall	Forward fall	21	84
	Backward fall	21	
	Leftward fall	21	
	Rightward fall	21	
ADLs	Standing	378	1940
	Walking	769	
	Walking upstairs	262	
	Walking downstairs	303	
	Standing up and sitting down	228	

large datasets (i.e., thousands or more training samples), but for small datasets, the quasi-Newtonian optimizer lbfgs can converge faster and perform better, and alpha can be used to control possible "overfitting" and "underfitting." In this paper, the solver is set to lbfgs. The three-layer BP neural network not only achieves a good detection effect but also does not compromise the lead time by making the operations too complex.

(2) KNN: KNN is a common classification and regression method. The classification algorithm used to build the KNN model is KNeighborsClassifier. The parameters that can be set in the algorithm include K value, distance measure, weight, and algorithm. Among them, the K value is the most important parameter. If the K value is too large, it means that the model complexity is low, which may lead to underfitting. If the K value is too small, it is easy for the model to be sensitive, which may lead to overfitting. Therefore, the grid search algorithm in the Scikit-learn library is used to obtain the K value in combination with cross-validation, and the final parameter is confirmed to be 3. The distance metric is set to the default value of "minkowski." The weight is set to the default value of uniform, which means that each region has the same weight. The classification strategy is set to auto, which will automatically select an appropriate strategy based on the input samples.

(3) SVM: SVM is a commonly used supervised learning method, which improves the generalization ability of the algorithm by minimizing structural risks, and is widely used in statistical classification and regression analysis. The SVM algorithm library in Scikit-learn is divided into two categories; one is the classification algorithm library, including SVC, NuSVC, and LinearSVC3 classes. The other is the regression algorithm library, including SVR, NuSVR, and LinearSVR3 classes. To solve the classification problem, the SVM algorithm model in this paper chooses the SVC classification algorithm because it cannot determine whether the data can be linearly fitted. The core parameters of the SVM model are the regularization parameters c, tol, and kernel. The strength of the regularization is inversely proportional to C. It must be strictly positive. The regularization parameter c in this article also uses the grid search algorithm provided by Scikit-learn combined with cross-validation to obtain the optimal value and finally obtains the optimal model parameter $c = 1$. The tolerance tol of the kernel function kernel and the stopping criterion are set to the default values.

(4) RF: random forest is a highly flexible machine learning algorithm with considerable promise for a wide range of applications. When training the random forest model, the random forest classifier function in the Scikit-learn library is specifically called. Since this algorithm itself is not sensitive to the setting of hyperparameters, it also serves to simplify the model. The hyperparameter that this paper decides to set is the number of decision trees in the forest, i.e., the maximum number of iterations n_estimators of the classifier. By using grid search and tenfold cross-validation method, the optimal parameter value is 10.

In this study, to evaluate the performance of each algorithm, the results were analyzed using statistical parameters (sensitivity, specificity, accuracy, and lead time). The fall detection results are generally divided into four situations, which are expressed in this system as.

- True positive (TP): a fall occurs, and the system detects it.
- False positive (FP): an ADL occurs, but the system detects a fall.
- True negative (TN): an ADL occurs, and the system detects it.
- False negative (FN): a fall occurs, but the system detects an ADL.

This work aims to evaluate the performance of each classifier using the statistical parameters calculated from Eqs. (5)–(8). Accuracy represents the average classification ability of the system; sensitivity is the ability of the algorithm to detect falls, with higher sensitivity representing a lower rate of missed falls. Specificity is the ability of the algorithm to detect ADLs. The lead time (T_{lead}) is the time between when the system detects a fall and when it touches the ground after a fall has occurred, i.e., the reaction time reserved for fall protection. In Eq. (7), $t_{\text{dectecting}}$ indicates the algorithm's computation time, $t_{\text{WIFI_delay}}$ denotes the WIFI data transmission delay, and $t_{\text{filtering}}$ is the sensor's filtering time.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}} \tag{5}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{6}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{7}$$

$$T_{lead} = \left(t_{impact} - t_{detected}\right) - t_{dectecting} - t_{WIFI_delay} - t_{filtering} \tag{8}$$

To evaluate the performance of the system more intuitively, a confusion matrix is proposed to be used for the representation. The entire sample dataset is divided into 4:1 training set and test set, and the data from the training set was then used to train the four algorithms to obtain the algorithm models. The algorithmic model derives the prediction results from the test dataset and then represents the prediction result with the true value in a confusion matrix. Each column of the confusion matrix represents the prediction result of the model for the input samples, and the total number of each column represents the number of samples predicted to be the category, each row represents the true category of the sample, and the total number of each row indicates the number of input samples in that category. The confusion matrix for the four algorithms is shown in Fig. 6, where the contents in the square start from the top left to the bottom right, indicating true positive (TP), false positive (FP), true negative (TN), and false negative (FN) in that order.

Through the confusion matrix, the sensitivity, specificity, and accuracy of each algorithm can be calculated, subsequently, by inputting 100 datasets for each algorithm and recording the total running time of the program, and then finding its average value $T_{dectecting}$ as the algorithm operation time (including the data preprocessing time). The algorithm $T_{detected}$ is related to the actual position detected by the algorithm, and finally, the lead time of the detection system can be calculated according to Eq. (8).

The accuracy of all four algorithms shown in Table 2 was excellent, with the highest accuracy being 99.9% for RF and BP neural networks and the worst performance of 96% for SVM, indicating that each classifier was robust in recognizing falls. The lead time of KNN and RF was found to be the longest, with a lead time of 460.95 ms. In addition, all four algorithms were calculated to have lead times greater than 400 ms, all of which meet the lead time requirements set for this system. However, in order to obtain the best algorithm, performance analysis of the classifiers alone is not sufficient, and further analysis of the algorithms' ability to fit the available data is required.

3.3 Fitting ability

To evaluate the performance of the classifier, receiver operating characteristic (ROC) curves and learning curve analysis are also provided for the four classification

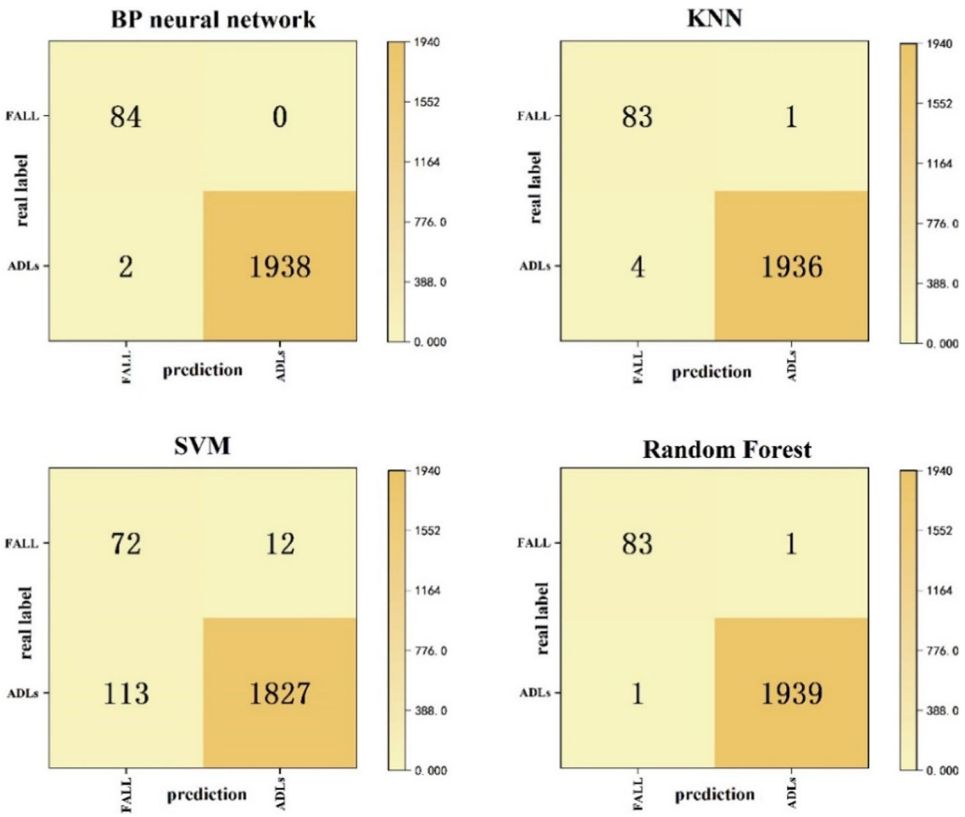


Fig. 6 Confusion matrix for the four algorithms

systems. ROC is a composite indicator reflecting a continuous variable of sensitivity and specificity, the vertical coordinate of the curve is the true positive rate (TPR), which is the number of true positive samples detected divided by the number of true positive samples, and the horizontal coordinate is the false positive rate (FPR), which represents the number of false positive samples detected divided by the number of all true negative samples. The evaluation of algorithm performance based on ROC alone is not comprehensive and another concept, area under the curve (AUC), needs to be introduced. The AUC value, as a quantitative value, allows for quantitative comparisons, and in this study, the AUC value is reported in the legend of the ROC curve. The following criteria are used to determine the AUC value:

(1) AUC≈1.0: optimal outcome scenario
(2) AUC between 0.7 and 0.9: higher detection accuracy
(3) AUC between 0.5 and 0.7: lower detection accuracy
(4) AUC below 0.5: the detection algorithm is useless

The learning curve is used to see how the model performs on new data and to determine whether the model has high variance bias or high bias and whether the overfitting of the model decreases as the training sample increases. By using the tenfold cross-validation method [47–49], the input data is randomly divided into 10 parts, and 9 of these parts are alternately reserved for training and one-tenth for testing. After 10 rotations in this way, the mean and variance of the test scores are taken, and then, the process is repeated by continuously increasing the sample data. Finally, a complete learning curve is plotted with the test scores as the ordinate and the number of sample data as the abscissa. The learning curve is generally divided into three cases:

(1) High deviation: when the scores of both the training and validation sets converge and are very low.
(2) High variance: when there is a large difference between the scores of the training and validation sets.
(3) Ideal situation: the deviation and variance are minimal, indicating that the algorithm fits the existing data very well.

Figure 7 shows the ROC and learning curve of the evaluated classification models. The results based on the ROC and AUC on the training dataset and the test dataset show that KNN and RF perform better than BP neural network and SVM for classification. Based on the convergence of the learning curve with increasing number of samples on the training (red line) and cross-validation (green line) sets, KNN fits the new dataset better. The learning curve of the RF converges to 1 as the dataset increases, but is nonlinear, making it more suitable for training sets with smaller amounts of data.

Table 2 Algorithm performance comparison

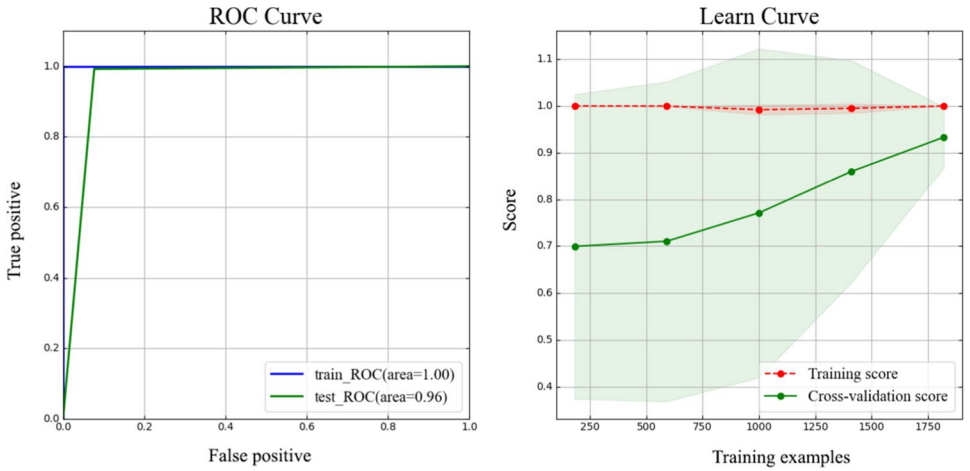
Network model	Accuracy	Sensitivity	Specificity	Lead time
BP neural network	99.9%	100%	99.8%	455.85 ms
KNN	99.7%	98.8%	99.8%	460.95 ms
SVM	96.0%	85.7%	94.2%	450.80 ms
RF	99.9%	98.8%	99.9%	460.95 ms

3.4 Results

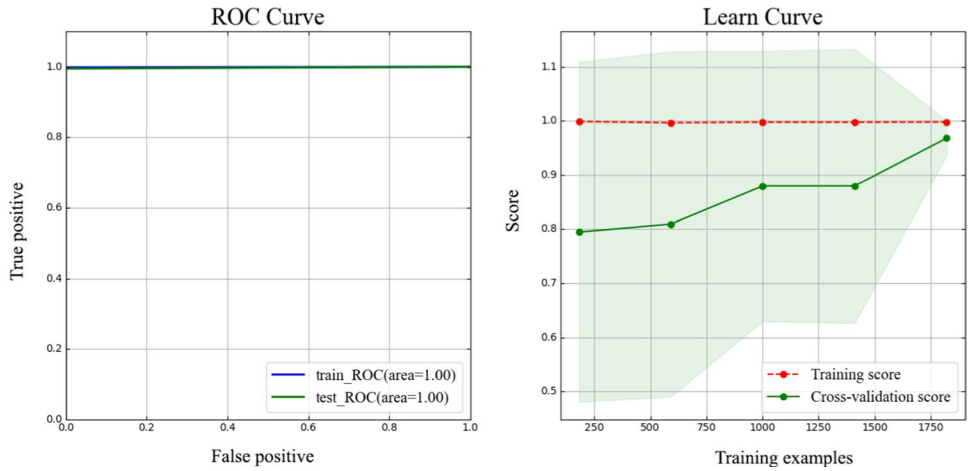
Based on the results of the performance and fitting ability of the four fall detection algorithms, the following conclusions can be drawn. The BP neural network has good accuracy and sensitivity with its test ROC curve of 0.93. However, the learning curve of the BP neural network shows that it has a better fitting ability only when the amount of data is more significant than 1600. The KNN algorithm shows a high detection ability, and the algorithm model fits the available data very well, with excellent accuracy as demonstrated by its ROC curve with test roc = 1; it also has the longest lead time among the four algorithms. However, its fitting ability does not change when the data volume is in the range of 1000–1400, so when the data volume is in this range, we might choose another algorithm or choose to increase the data volume. The SVM has the weakest fitting ability among the four algorithms, with an AUC value of only 0.92 for its test dataset, which indicates that it has the worst ability to detect falls compared to the other algorithms. The learning curve results show that the initial fit of the RF is not ideal, but the curves of the training and cross-validation sets move closer together and gradually converge to a score close to 1 as the amount of sample data increases for data sizes greater than 1000.

Since KNN and BP neural networks were used as the detection algorithms for this system, the RF algorithm was selected for training when the amount of data was greater than 1000 to derive the pre-impact fall detection model, and the KNN algorithm was selected when the amount of data was smaller, with the two algorithms switchable in the software.

In the final stage, the system work for pre-impact detection was largely completed by selecting the ideally performing KNN and BP neural networks to be embedded in the host computer algorithm, in order to further demonstrate the detection performance and usability of this system. As shown in Fig. 8, we conducted out a validation experiment in a real environment where a subject wearing a pre-impact fall detection device fell while walking, and the results were updated in real time by the test on display interface of the host computer, and the results responded accurately when the fall occurred.



(a) BP neural network



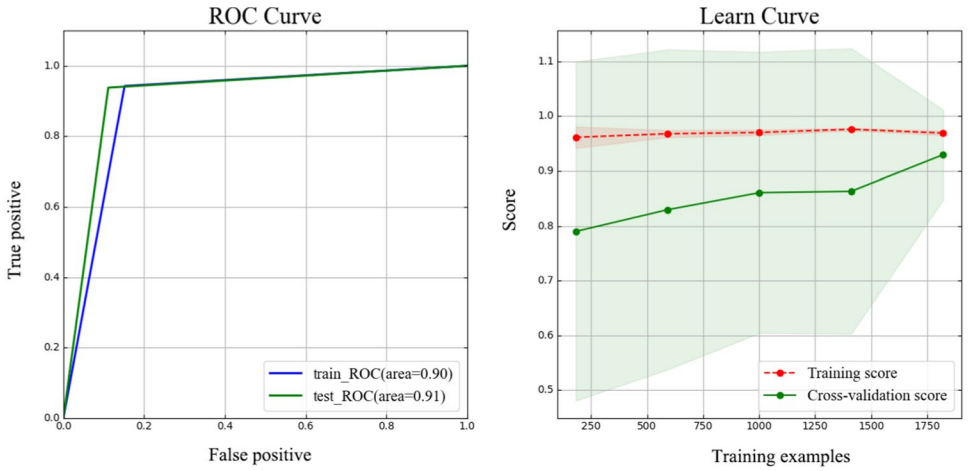
(b) KNN

Fig. 7 ROC curves (left) and learn curves (right) of the four algorithms

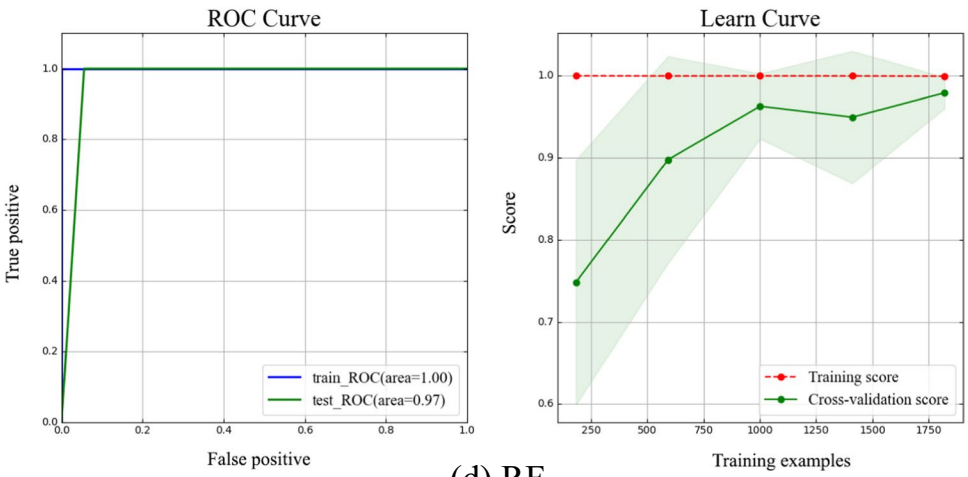
In addition, the delay of WIFI at a distance of 10 m t_{WIFI_delay} is about 1 ms, the filtering time of the sensor $t_{filtering}$ is about 37.6 ms, and the other elapsed time is almost negligible. Therefore, the data acquisition time is $t_{filtering} + t_{WIFI_delay} = 38.6$ ms. By inputting 100 sets of data to the KNN and BP neural network algorithms selected in the article, the total running time of the program is recorded and then averaged as the response time of the host computer (including the time of data preprocessing), the response time of the KNN algorithm is 0.45 ms, and the response time of the BP neural network is 0.55 ms.

Table 3 shows the comparative research in the literature on fall detection based on pre-impact sensors over the last 5 years. The research by Wu et al. is based on the MPU6050 installed in the waist and thigh for fall detection, using the Fisher classifier to achieve a lead time of 376–404 ms. Ahn et al. also used the MPU6050 inertial sensor, the sensor position of which is only the waist and the threshold method based on the vertical angles and triangle feature for fall detection, and achieved 100% sensitivity, but the specificity value was not ideal. Shi et al. achieved a longer lead time of 400 ms, using a class activation mapping method based on CNN for detection. Xiao et al. performed fall detection based on SVM with surface electromyographic (sEMG) signal of leg muscles, and the lead time was not ideal. Zhong et al. employed a threshold based on a vertical velocity method and achieved a lead time of 363 ms. In this study, we propose a pre-impact fall detection method based on multisensors installed on the foot, using the KNN and BP neural network algorithms with the best fall detection performance for identification. The achieved KNN had 98.8% sensitivity, 99.8% specificity, and 99.7% accuracy, and BP neural network had 100% sensitivity, 99.8% specificity, and 99.9% accuracy, and their lead times were both 460.95 ms. Lead time and other aspects are better than the current literature, and the specificity is 100% as in the research of Ahn et al.

Fig. 7 (continued)



(C) SVM



(d) RF

3.5 Power consumption

The STM32F103C8T6 is in the run mode in the idle state; when it is not connected to any other components, the operating voltage is 3.3 V; it consumes about 36 mA. The FSR402 is a low-power thin-film pressure sensor with an operating voltage of 5 V, a resistance range of tens of ohms to about 65 MΩ, and a power consumption of about 20 mW.

Fig. 8 User prototypes in a real-time environment



Table 3 Performance comparison of the proposed method with state-of-the-art pre-impact fall detection

Ref	Sensors and position	Methods and basic principles	Sensitivity (%)	Specificity (%)	Lead time (ms)
Wu [35] (2019)	MPU6050, waist and thigh	A hierarchical classifier (Fisher)	95.5	97.3	376–404
Ahn [13] (2018)	MPU9150, waist	Threshold based on vertical angles (VA) and triangle feature (TF)	**100**	78.3 (VA) 83.9 (TF)	401 (VA) 427 (TF)
Shi [31] (2020)	MPU9250, waist	A class activation mapping (CAM) method based on CNN	94.4	97.17	400
Xiao [50] (2018)	sEMG, legs	SVM	93.71	92.67	202.4
Zhong [34] (2018)	IMU_JY61P, waist	Threshold based on a vertical velocity	93.6	95.6	363
Ours	MPU-6050, FRS402, foot	KNN, BP neural network	98.8 (KNN), **100** (BP neural network)	**99.8**	**460.95 ms**

Values in bold indicate the performance of the design's pre-impact fall detection system

The operating voltage of the MPU6050 inertial sensor module is 3–5 V, and the power consumption is about 2.2 mA when the sampling frequency is set to 1 kHz. The operating voltage of the ESP8266 chip is 3.3 V, and it supports three low-power sleep modes. In deep sleep mode, the module maintains current at 10 μA; in light sleep mode, the module operates at 0.9 mA; in RF sleep mode, the module draws 15 mA. Therefore, the average power consumption of the STM32F103C8T6 with all components connected and operating is 100–120 mA, measured with a multimeter in series with the PCB, and it can run continuously for about 24 h with a 2800 mA power supply.

4 Discussion

4.1 System limitations and future work

In this study, a multisensor-based pre-impact fall detection system was developed that performed well overall. However, there are limitations to our approach. First of all, the detection range of the system is limited. Only four common fall directions were considered in this study, and other types of falls such as those caused by bumping, slipping, and tripping were not discussed. For the elderly, the reasons for falls may also vary from person to person, so how to detect different types of falls is also a problem that needs to be addressed. In addition, although the plantar pressure sensor can be positioned according to the size of the subject's foot, some invalid pressure data were inevitably generated during the experiment, and how to collect the plantar pressure data more reliably needs to be considered in future studies. Also, the adaptability of the system should be improved so that it can be applied to more environments and usage scenarios. Finally, in terms of increasing the usability of the system, a user-friendly interface and instruction manual can be developed so that users can better understand how to operate the system. In future work, we can focus on optimizing and improving the above aspects to enhance the performance and practicality of the fall detection system.

4.2 Implementation challenges

The fall detection system requires a large amount of historical data to train machine learning models. Due to the insufficient data in the public dataset, most of the fall data sources used in the system testing are from young people, which is somewhat different from the real-world fall data of the elderly. Currently, it is a challenge to collect a complete dataset of falls.

In addition, the system needs to run and collect data for a long time, so it needs to have enough battery capacity to support this. However, due to its small size, the space in which the battery can be charged is limited. Ensuring that it has sufficient range is another issue.

Moreover, the detection system may collect and store personal information, which may raise privacy concerns. Therefore, the design of the fall detection system should consider how to protect personal privacy.

4.3 System stability

Fall detection is very important to help the elderly and disabled avoid falls and ensure safety. However, the stability of the fall detection system has become a key challenge due to the complex problems faced by the fall detection system, such as the diversity of human movement patterns, the weak anti-interference ability of WIFI transmission, and the timely response.

First, different people fall in various ways, so a fall detection system must be able to identify many different fall patterns. The current study was able to determine four types of falls and five activities of daily living that meet daily needs. Second, exposure of the sensor to the environment is also an issue for fall detection systems. The fall detection system in this study needs to use the ESP8266 WIFI module, but this module is not very stable and is susceptible to noise such as vibration. In addition, a fall detection system should be able to respond to falls in time to intervene. The proposed pre-impact fall detection system has a lead time of up to 460. 95 ms, which can effectively prevent the impact of a fall if combined with safeguards.

5 Conclusion

In this paper, a new method of pre-impact fall detection based on wearable sensing devices is proposed and evaluated. The dataset is collected by the passive fall of the six-degree-of-freedom perturbation platform, and the data types include foot angular velocity, angular acceleration, and pressure sensor data. Features were extracted from the acquired sensor data, and data preprocessing was performed. The feature selection results show that the three types of sensor data from the foot are important for activity classification.

In this work, four commonly used machine learning algorithms (KNN, SVM, RF, and BP neural networks) were trained on the collected fall datasets and their performance was compared in terms of turnaround time, sensitivity and specificity, accuracy, and data fitting and learning capabilities. The training results showed that the KNN and BP neural networks performed well, with 98.8% sensitivity, 99.8% specificity, and 99.7% accuracy for KNN and 100% sensitivity, 99.8% specificity, and 99.9% accuracy for BP neural network. In addition, their lead times are both 460.95 ms and the fitting ability of KNN is the most excellent. Therefore, these two algorithms are built into the host computer interface and can be switched manually. Finally, the validation experiments show that the performance of this system fully meets the expectations, with the advantages of high accuracy and long lead time, which is a promising method for pre-crash fall detection.

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Declarations

Conflict of interest The authors declare no competing interests.

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